

Software Requirements Specification (SRS)

Online Internship & Placement Preparation Portal

Version: 1.0

Team No: 14

Course: Software Engineering

Date: 09-09-2026

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Revision History

Version	Date	Author	Change summary	Approval
1.0	09-09-2026	Team 14	Initial draft of the SRS for the Online Internship & Placement Preparation Portal	

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document lays out the software requirements for the Online Internship & Placement Preparation Portal, a web application meant to help students of the college prepare for and apply to internships and placement drives that are routed through the Training and Placement (T&P) cell. It is written for our own reference during design and development, and also so that whoever evaluates the project can see what the system is supposed to do before looking at the implementation.

1.2 Scope

The portal lets a student create a profile, build a resume, look at internship and placement drives that they are eligible for, apply to them, take practice aptitude and technical tests, prepare for interviews using a shared question bank, and track the status of their applications. On the other side, the T&P Officer (TPO) can post drives, set eligibility rules, update statuses, schedule mock interviews, and pull reports.

What it does not do: it does not process salaries or offer letters once a student is actually hired, it does not replace the official placement records the institution already maintains, and it is not meant to be a full HRMS. Those are handled outside the system.

1.3 Audience

This document is intended for the development team (that's us), the course instructor/evaluator, the T&P cell staff who would eventually use the admin side, and any student reviewer of the project.

1.4 Definitions, Acronyms and Abbreviations

- TPO – Training & Placement Officer
- FR – Functional Requirement
- NFR – Non-Functional Requirement
- UI – User Interface
- API – Application Programming Interface
- OTP – One-Time Password
- JWT – JSON Web Token, used for session authentication
- RTM – Requirements Traceability Matrix
- CGPA – Cumulative Grade Point Average, used as one of the eligibility filters

2. Overall Description

2.1 Product Perspective

The portal is a standalone web application — it is not replacing or plugging into the college's existing ERP, though later it could be extended to pull student academic records from there instead of asking students to enter their CGPA and backlog count manually. For this version, all data (student profiles, drives, applications, test results) lives in the portal's own database.

2.2 Major Product Functions

- Student registration and login
- Profile creation and resume building
- Browsing internship and placement drives
- Applying to a drive
- Aptitude and technical mock tests with auto-grading
- Interview preparation resources and mock-interview booking
- Application status tracking and notifications
- TPO dashboard for posting drives, verifying eligibility and viewing reports

2.3 User Roles and Characteristics

Student — the main user of the system. Assumed to have basic familiarity with using web forms and uploading files, but not much patience for a confusing interface, so the flows need to be short and obvious.

TPO / Admin — manages drives and eligibility from the back end. Spends more time on the system per session than a student does, so the dashboard needs to show a lot of information at a glance (applicant counts, drive status, etc.).

Recruiter — a limited, read-only role that can view the shortlisted candidates for a drive they posted. This role is a smaller part of the system and is mainly included so the TPO doesn't have to forward candidate lists manually.

2.4 Operating Environment

The application is a responsive web app, used through a browser (Chrome, Firefox or Edge) on a laptop, desktop, or phone. The backend runs on a cloud-hosted server with a relational database. No special hardware is required beyond a device with a browser and, for the mock-interview feature, an optional webcam.

2.5 Constraints

- Student sign-up is restricted to the institution's email domain, so an outsider can't register.
- The interface should stay usable on the campus wifi, which can be slow during peak hours — so pages should not depend on large uploads/downloads unless the student is explicitly uploading a resume or certificate.
- The mock-test engine should not depend on a paid third-party service, since this is a student project with no budget for licensing.

3. External Interface Requirements

3.1 User Interfaces

Two separate dashboards — one for students and one for the TPO — each with a simple sidebar/menu layout. Fonts and colour contrast follow basic accessibility guidelines so the pages stay readable, and pages are navigable with just a keyboard for anyone who needs it.

3.2 Hardware Interfaces

No dedicated hardware is required. A device with a browser is enough; a webcam and microphone are only needed if a student chooses to use the mock-interview video feature, and even that can fall back to an audio-only or text-based session if a webcam isn't available.

3.3 Software Interfaces

- Email service (SMTP/API) for OTPs, password resets and drive notifications
- Cloud object storage for resumes, certificates and profile pictures
- Optional integration point for pulling additional internship listings from an external job-posting API (kept as a future enhancement, not built in v1.0)

3.4 Communications

All traffic between the browser and the server goes over HTTPS/TLS 1.2 or above. Sessions are handled using signed JWT tokens with a defined expiry, so a stolen token can't be used indefinitely.

4. System Features

Each feature below lists its functional requirements with an ID, acceptance criteria and a reference test case, in the same format the course template uses. IDs follow OIPP-F-####.

4.1 Registration & Authentication

Description: lets a student create an account and get back into it safely.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-001	The system shall allow a student to register using their institution email ID and verify it through an OTP sent to that email.	Functional	High	TPO / Student	AC-001: Registration completes only after correct OTP is entered within 5 minutes. Test: TC-Reg-01	OTP expires after 5 min
OIPP-F-002	The system shall allow a registered student to log in using their email and password.	Functional	High	Student	AC-002: Valid credentials open the dashboard; invalid ones show an error without stating which field is wrong. Test: TC-Auth-01	5 wrong attempts locks the account (see OIPP-SR-004)
OIPP-F-003	The system shall allow a student to reset a forgotten password through a time-limited link sent to their registered email.	Functional	Medium	Student	AC-003: Reset link works once and expires after 30 minutes. Test: TC-Auth-02	Old sessions invalidated after reset

4.2 Profile & Resume Builder

Description: lets a student maintain the information that drives typically ask for, once, instead of retyping it for every application.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-004	The system shall allow a student to create and edit an academic and personal profile (branch, CGPA, backlog count, skills, contact details).	Functional	High	Student	AC-004: Saved profile fields persist and are shown correctly on re-login. Test: TC-Prof-01	Feeds eligibility checks in F-008
OIPP-F-005	The system shall allow a student to upload a resume (PDF, max 5 MB) and download it back at any time.	Functional	High	Student	AC-005: Uploaded PDF is retrievable and opens correctly. Test: TC-Prof-02	Non-PDF or oversized files rejected with a message
OIPP-F-006	The system shall provide a resume template that auto-fills from the student's profile data, which the student can then adjust before saving.	Functional	Low	Student	AC-006: Auto-filled template matches the latest saved profile fields. Test: TC-Prof-03	Nice-to-have; not blocking for v1.0

4.3 Placement / Internship Drives

Description: covers posting a drive, filtering it to eligible students, and applying.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-007	The TPO shall be able to post a new drive with company name, role, package/stipend, eligibility criteria and application deadline.	Functional	High	TPO	AC-007: A posted drive appears in the drive list within the same session. Test: TC-Drv-01	
OIPP-F-008	The system shall show a student only the drives for which they meet the stated eligibility criteria (CGPA, branch, backlog count).	Functional	High	TPO / Student	AC-008: A student below the CGPA cut-off does not see that drive in their list. Test: TC-Drv-02	Depends on OIPP-F-004 profile data being accurate
OIPP-F-009	The system shall allow an eligible student to apply to a drive using their stored profile and resume, without re-entering information.	Functional	High	Student	AC-009: Application is recorded and visible under 'My Applications'. Test: TC-Drv-03	
OIPP-F-010	The system shall prevent a student from applying to the same drive more than once.	Functional	Medium	TPO	AC-010: Second attempt shows 'already applied' instead of creating a duplicate record. Test: TC-Drv-04	

4.4 Mock Tests & Aptitude Practice

Description: lets a student practice aptitude and technical questions before the real tests that come with a drive.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-011	The system shall provide topic-wise aptitude and technical mock tests with a timer.	Functional	High	Student	AC-011: Test auto-submits when the timer runs out. Test: TC-Test-01	
OIPP-F-012	The system shall auto-grade objective mock tests and show the score immediately after submission.	Functional	High	Student	AC-012: Score shown matches the number of correct answers. Test: TC-Test-02	
OIPP-F-013	The system shall maintain a history of a student's past mock test attempts and scores.	Functional	Medium	Student	AC-013: Past attempts are listed in chronological order under the student's profile. Test: TC-Test-03	

4.5 Interview Preparation

Description: gives a student access to company-wise interview questions and a way to get mock-interview practice with feedback.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-014	The system shall provide a searchable bank of previously-asked interview questions, tagged by company.	Functional	Medium	Student	AC-014: Searching by company name returns only that company's tagged questions. Test: TC-Int-01	
OIPP-F-015	The system shall allow a student to book a slot for a peer or senior mock interview.	Functional	Medium	Student	AC-015: A booked slot appears on both the student's and the reviewer's schedule. Test: TC-Int-02	
OIPP-F-016	The system shall allow feedback to be recorded against a mock interview session after it is completed.	Functional	Low	Student	AC-016: Feedback text is saved and visible when the student reopens that session. Test: TC-Int-03	

4.6 Application Tracking & Notifications

Description: keeps a student informed about where their applications stand and about new drives or deadlines.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-017	The system shall show an application's status (applied / shortlisted / rejected / selected) as updated by the TPO.	Functional	High	Student / TPO	AC-017: Status change made by the TPO reflects on the student's dashboard without delay. Test: TC-Trk-01	
OIPP-F-018	The system shall send an email or in-app notification when a new drive is posted or a deadline is approaching.	Functional	Medium	Student	AC-018: Notification is received within a few minutes of the triggering event. Test: TC-Trk-02	

4.7 TPO Dashboard & Reporting

Description: gives the TPO a quick view of how placements are progressing and a way to pull data out for reporting.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-F-019	The TPO shall be able to view placement statistics (number of applications, selections, package range) on a dashboard.	Functional	Medium	TPO	AC-019: Dashboard figures match the underlying application records. Test: TC-TPO-01	
OIPP-F-020	The system shall allow the TPO to export the applicant list for a drive as a CSV file.	Functional	Low	TPO	AC-020: Exported CSV contains one row per applicant with the expected columns. Test: TC-TPO-02	

5. Non-Functional Requirements

These are measurable requirements tied to the acceptance tests listed in Section 6. IDs follow OIPP-NF-####.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-NF-001	Page loads and test-submission responses shall complete within 3 seconds for 90% of requests under normal load.	Performance	High	Team	90th percentile response time ≤ 3 s measured under simulated load. Test: TC-Perf-01	
OIPP-NF-002	The portal shall be available at least 99% of the time in a month, excluding announced maintenance windows.	Reliability	High	Team	Uptime log shows $\geq 99\%$ monthly. Test: server logs review	
OIPP-NF-003	Passwords and other sensitive fields shall never be stored or logged in plain text.	Security	High	Team	Database inspection shows only hashed values. Test: TC-Sec-01	See OIPP-SR-002
OIPP-NF-004	The portal's pages shall remain usable on a screen reader and maintain readable contrast, broadly following WCAG 2.1 AA.	Usability	Medium	Team	Manual accessibility check passes on key pages. Test: TC-UX-01	
OIPP-NF-005	The system shall support at least 2,000 concurrent students without significant slowdown during peak application windows (e.g., deadline day).	Scalability	Medium	Team	Load test with 2,000 simulated users shows response times within NF-001 bounds. Test: TC-Perf-02	

5.1 Security

5.1.1 Security Objectives

- Protect student personal and academic data (profile, resume, marks, contact details) from unauthorized access or leakage.
- Make sure only verified, eligible students can view and apply to a given drive, and that only authorized TPO/admin accounts can post drives or change eligibility rules.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test ref	Comments / Dependencies
OIPP-SR-001	All communication between the browser and the server shall use HTTPS with TLS 1.2 or higher.	Security	High	Team	AC-SR-001: Non-HTTPS requests are redirected/rejected. Test: TC-Sec-02	
OIPP-SR-002	Passwords shall be stored using a salted hash (e.g. bcrypt) and never in plain text or reversible encryption.	Security	High	Team	AC-SR-002: Database dump shows only hashes. Test: TC-Sec-01	
OIPP-SR-003	Student registration shall be restricted to verified institution email addresses.	Security	High	TPO	AC-SR-003: Sign-up with a non-institution email is rejected. Test: TC-Sec-03	
OIPP-SR-004	The system shall lock a student account for 15 minutes after 5 consecutive failed login attempts.	Security	Medium	Team	AC-SR-004: 6th attempt is blocked with a lockout message. Test: TC-Sec-04	
OIPP-SR-005	Only accounts with the TPO/admin role shall be authorized to post, edit or delete a drive, or change eligibility criteria; every such action shall be checked against the account's role before it is carried out.	Security	High	TPO	AC-SR-005: A student account attempting these actions via the API is rejected with a 403. Test: TC-Sec-05	Role-based access control (RBAC)

6. Quality Attributes & Acceptance Tests

Exit criteria: all high-priority functional requirements are implemented and pass their acceptance test, no critical NFR fails during testing, and the RTM in Section 8 shows every listed test case passed.

Acceptance test suites planned:

- Registration & Authentication
- Profile & Resume
- Drive Posting & Application
- Mock Test Engine
- Interview Preparation & Booking
- Notifications
- TPO Dashboard & Reporting
- Security (access control, password storage, session handling)

7. UML Use-Case Diagrams

Two use-case diagrams cover the two main actor groups: the student, who is the primary user, and the TPO/Admin (with the recruiter as a secondary actor with limited access).

7.1 Student — Use-Case Diagram

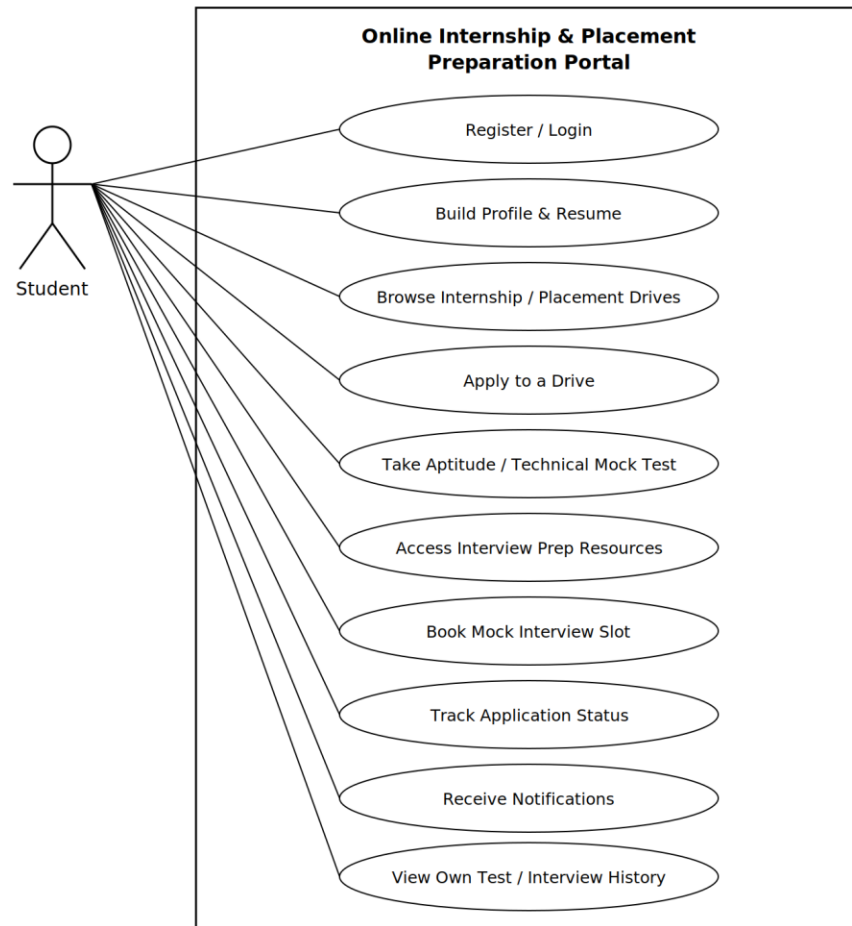


Figure 1: Use cases available to a student

7.2 TPO / Admin — Use-Case Diagram

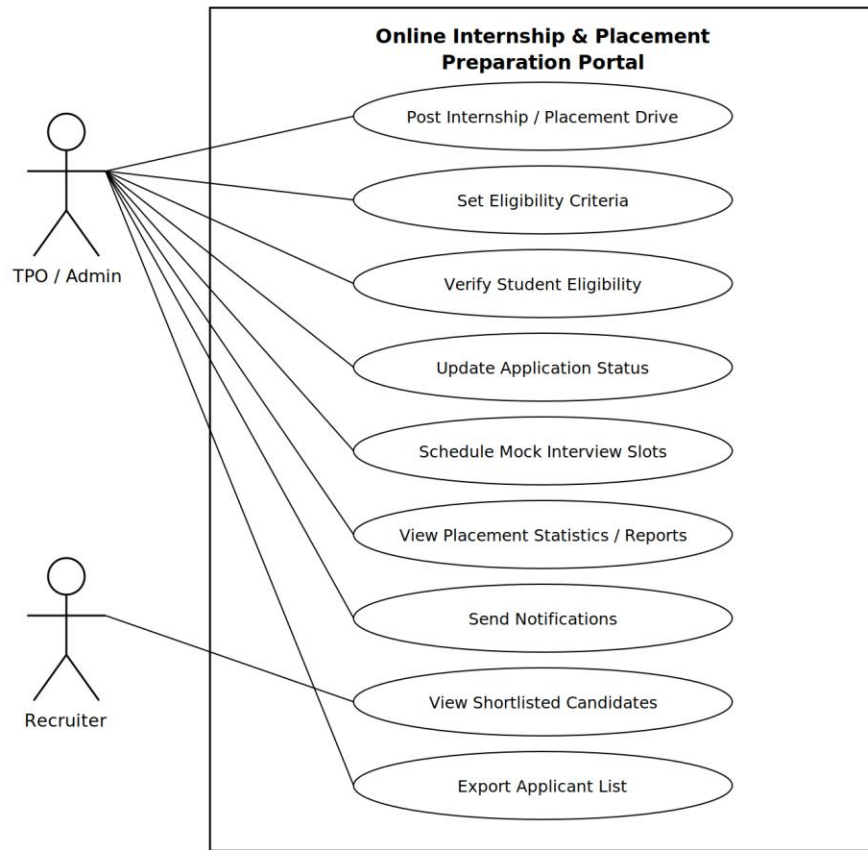


Figure 2: Use cases available to the TPO/Admin, with the Recruiter as a secondary actor

8. Requirements Traceability Matrix (RTM)

A sample of the RTM is shown below; the full matrix covering every requirement ID will be maintained alongside the test plan as development progresses. Status: N = Not tested, P = Passed, F = Failed, A = Automated.

Req ID	Requirement (short)	Design Spec	Module	Test case(s)	Status	Comments
OIPP-F-001	Register & verify via OTP	4.1 / DS-Auth-01	AuthModule	TC-Reg-01	P	
OIPP-F-002	Login	4.1 / DS-Auth-02	AuthModule	TC-Auth-01	P	
OIPP-F-007	Post a drive	4.3 / DS-Drive-01	DriveModule	TC-Drv-01	N	Not yet implemented
OIPP-F-008	Filter drives by eligibility	4.3 / DS-Drive-02	DriveModule	TC-Drv-02	N	
OIPP-F-009	Apply to a drive	4.3 / DS-Drive-03	DriveModule	TC-Drv-03	N	
OIPP-F-011	Take a mock test	4.4 / DS-Test-01	TestModule	TC-Test-01	N	
OIPP-NF-001	Response time target	5 / DS-Perf-01	Web UI / API	TC-Perf-01	N	
OIPP-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-01	N	